SANDEEP KUMAR K

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Portfolio: sandeepk1028.github.io

PROFESSIONAL SUMMARY

Embedded Engineer with 2+ years of experience in firmware development, board bring-up and system validation. Core expertise in Bluetooth, BLE Audio and Auracast transmitter and receiver on STM32WBA55G and NXH3675, scaling I2S/TDM audio from mono to multi-channel. Proficient in Embedded C, Python, UART, SPI and I2C, ASPICE-aligned development, and lab debugging with oscilloscopes and logic analyzers.

PROFESSIONAL EXPERIENCE

OpenEcore Private Limited — *April 2025 – Present*

(formerly Kruthak Technology Private Limited)

Embedded Engineer — *October 2026 – Present*

- Built Auracast transmitter, receiver and assistant devices, and scaled I2S/TDM audio from mono to 4-channel using STM32WBA55G and NXH3675.
- Developed SkyAmour, an Android app that enables Auracast on devices without native Auracast support.

Associate Embedded Engineer — *April 2025 – October 2026*

- Developed embedded hardware and firmware for COFRO, a coffee bean plucking machine, moving the process from manual work to a semi-automated system.

Factana Computing Private Limited — *November 2023 – July 2024*

Junior Embedded Engineer

- Hardware, firmware and validation for three commercial device lines (APM, DWK, OCR), including a custom RS232 test kit.

Pi Square Technologies — *February 2023 – June 2023*

Embedded Engineer Intern

- Developed a sensor-based alcohol and drowsiness detection system with automatic safety actions.

KEY PROJECTS

- **COFRO & AURACAST:** Auracast transmitter, receiver and assistant; coffee bean plucking machine.
- **SkyAmour:** Android app that enables Auracast on devices without native Auracast support.
- **APM, OCR & DWK Devices:** Firmware, hardware and validation for three commercial device lines.
- **Alcohol & Drowsiness Detection System:** Driver safety system with alcohol and drowsiness sensors.
- **Smart Gardener:** Autonomous lawn-cutting robot with sensor-based terrain navigation.

TECHNICAL SKILLS

Programming Languages: C, Embedded C, Python
Microcontrollers & MPUs: STM32 (STM32WBA55G), NXP (NXH3675), ESP32, Raspberry Pi, Arduino
RTOS & IDEs / SDKs: FreeRTOS, STM32CubeIDE, MCUXpresso, ESP-IDF
Protocols & Communication: UART, I2C, SPI, RS232, I2S, TDM, Bluetooth Classic, BLE Audio, Auracast, LC3 Codec, IoT Data Transfer
Concepts & Tools: Embedded Linux, Hardware-Software Integration, ASPICE, System Validation, Schematic Design, PCB Layout, BOM Preparation
Debugging & Testing: Oscilloscope (CRO), Logic Analyzer, Multimeter, Protocol Analyzers, Serial Port Monitors, JTAG/SWD Debuggers

EDUCATION

B.E., Electronics & Communication — MVJ College of Engineering, Bengaluru — *2020 – 2023*
Diploma, Electronics & Communication — Govt. Polytechnic, Chintamani — *2017 – 2020*